

Discrete Isogonal Nets with Similar Parallelograms

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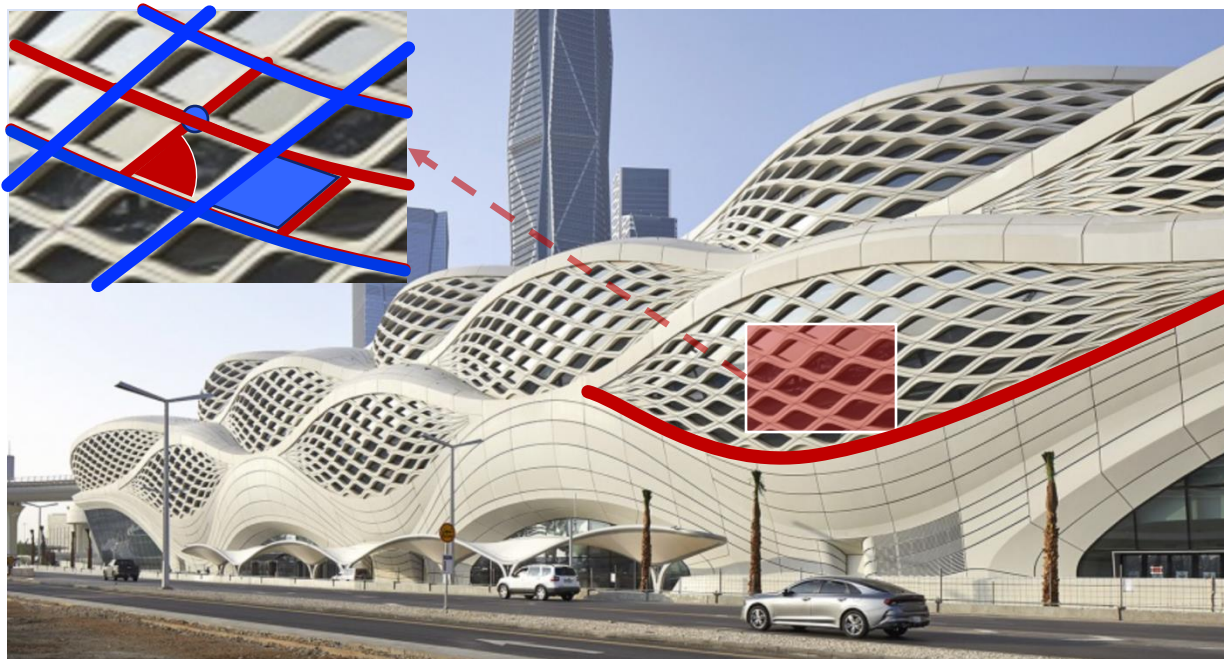


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UNIVERSITY

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University of East Anglia

Background

- High-quality surfaces are essential in architecture and product design



KAFD Metro Station in Riyadh, designed by ZHA

regular vertex

smooth curve

equal opposite sides

right angle

flat face

tessellation density

feature line alignment



Background

Orthogonal nets (O-nets)

- basic for other specialized quad nets, e.g. principal curvature net, minimal net,...
- offer aesthetic appeal and engineering practicality

Isogonal nets (I-nets)

- Constant curve angle beyond 90°
- Compelling alternative to O-nets
- Broader control over shape
- Unique visual effects
- Adaptable structural layouts



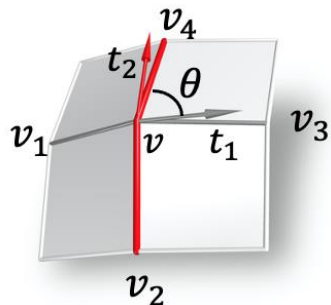
Yas Hotel, Abu Dhabi



Phoenix Media Center, Beijing



Overview of I-nets



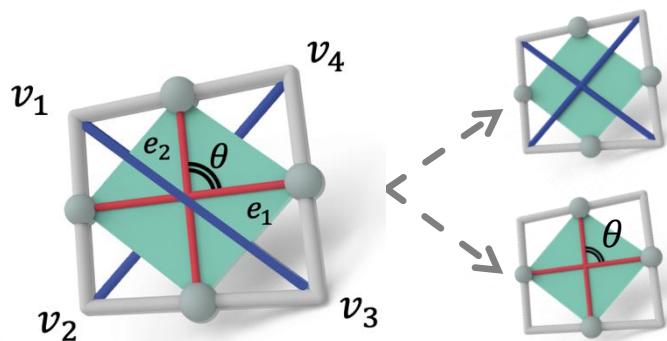
O-net: $\theta = \frac{\pi}{2} \Leftrightarrow (e_3 - e_1) \cdot (e_4 - e_2) = 0$

$e_i = \frac{v_i - v}{\|v_i - v\|}, (i = 1, 2, 3, 4)$

I-net: $\theta = \text{const.} \Leftrightarrow t_1 \cdot t_2 = \cos \theta$

$t_1 = \frac{e_3 - e_1}{\|e_3 - e_1\|}, t_2 = \frac{e_4 - e_2}{\|e_4 - e_2\|}$

Defined unit tangents $\dashrightarrow$ increase computation cost



Mid-edge subdivided quad is a parallelogram $\square$

O-net: $\theta = \frac{\pi}{2} \Leftrightarrow (v_3 - v_1)^2 = (v_4 - v_2)^2$

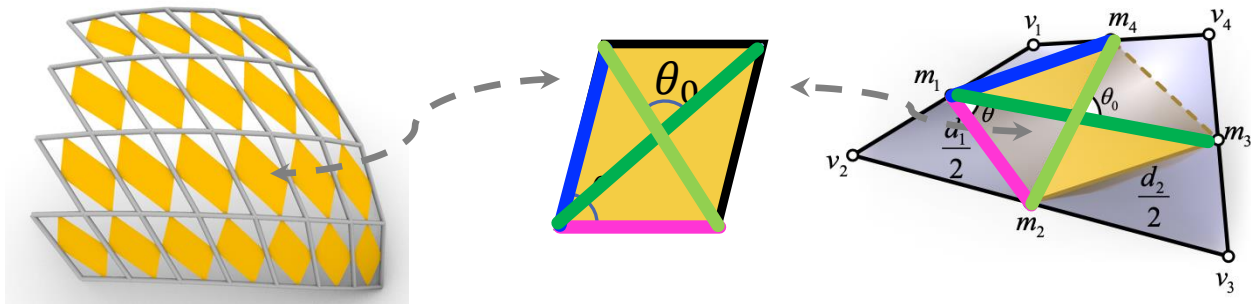
I-net: $\theta = \text{const.}$
 $\Leftrightarrow ?$



Checkerboard pattern (CBP) $\dashrightarrow$ How to represent I-net by edge lengths?

Proposed Method

- I-nets using similar mid-edge subdivided parallelograms



$$\frac{\|d_2\|}{\|d_1\|} = \lambda$$

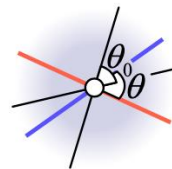
$$\frac{\|m_1 m_3\|}{\|m_2 m_4\|} = \mu$$

- Constraining edge ratios instead of angles

- Generalizations of O-nets

- Rhombus  • Rectangle  • Square 

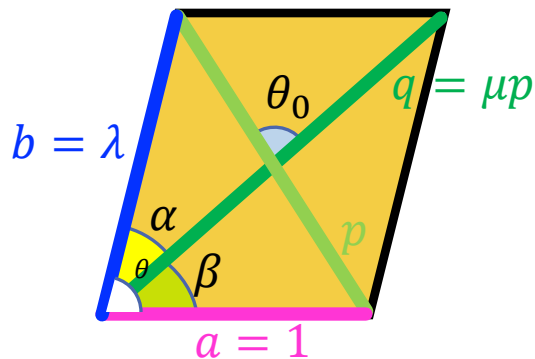
- Generalization to isogonal 4-webs (I-webs)



- Versatility of I-nets and I-webs in high-quality surface



Parallelogram Geometry

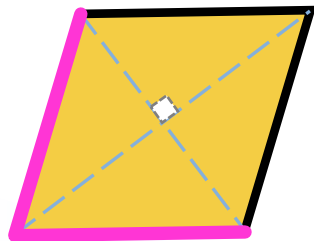


$$a : b : p : q = 1 : \lambda : \sqrt{\frac{2(1+\lambda^2)}{1+\mu^2}} : \mu \sqrt{\frac{2(1+\lambda^2)}{1+\mu^2}}$$

$$\cos \theta = \frac{(1 + \lambda^2)(\mu^2 - 1)}{2\lambda(1 + \mu^2)}, \cos \theta_0 = \frac{(1 + \mu^2)(\lambda^2 - 1)}{2\mu(1 + \lambda^2)}$$

$$\cos \alpha = \frac{3\lambda^2\mu^2 + \lambda^2 + \mu^2 - 1}{2\lambda\mu \sqrt{2(1 + \lambda^2)(1 + \mu^2)}}, \cos \beta = \frac{\lambda^2\mu^2 - \lambda^2 + 3\mu^2 + 1}{2\mu \sqrt{2(1 + \lambda^2)(1 + \mu^2)}}$$

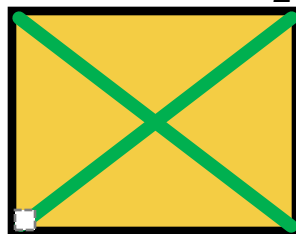
$$\lambda = 1 \Leftrightarrow \theta_0 = \frac{\pi}{2}$$



Rhombu

s

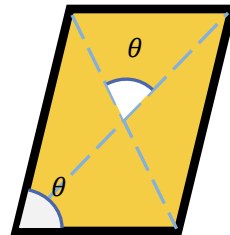
$$\mu = 1 \Leftrightarrow \theta = \frac{\pi}{2}$$



Rectangl

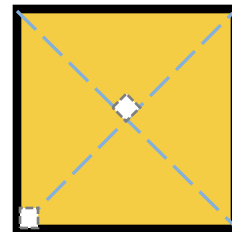
e

$$\lambda = \mu \Leftrightarrow \theta = \theta_0$$



θ -Parallelogram

$$\lambda = \mu = 1 \Leftrightarrow \theta = \theta_0 = \frac{\pi}{2}$$



Squar



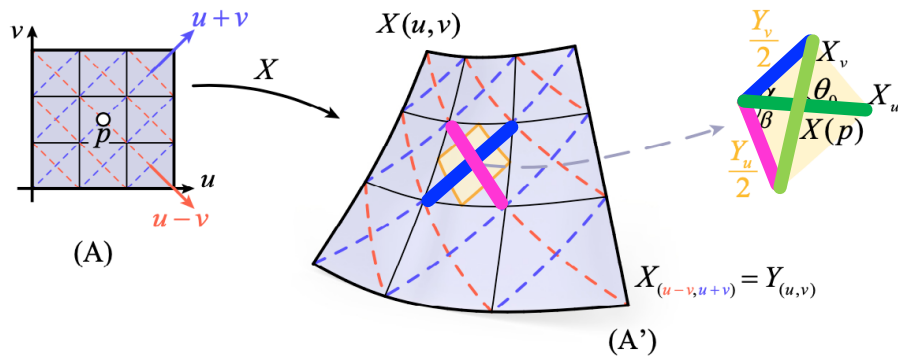
Isogonal Nets and Webs

Surface parametrization:

$$X: (u, v) \in \mathbb{R}^2 \rightarrow X(u, v) \in \mathbb{R}^3$$

Diagonal parametrization:

$$Y(u, v) = X(\textcolor{violet}{u} - \textcolor{violet}{v}, \textcolor{blue}{u} + \textcolor{blue}{v})$$



$$\begin{aligned} \text{constant } \frac{\|X_u\|}{\|X_v\|} &:= \mu \\ \text{constant } \frac{\|Y_v\|}{\|Y_u\|} &:= \lambda \end{aligned}$$

$\Rightarrow$

$$Y_v^2 = \lambda^2 Y_u^2 \Leftrightarrow (X_u + X_v)^2 = \lambda^2 (X_u - X_v)^2$$

$\Downarrow$

$$\theta(\lambda, \mu), \theta_0(\lambda, \mu), \alpha(\lambda, \mu), \beta(\lambda, \mu)$$

$$\cos \theta = \frac{(1 + \lambda^2)(\mu^2 - 1)}{2\lambda(1 + \mu^2)},$$

$$\cos \theta_0 = \frac{(1 + \mu^2)(\lambda^2 - 1)}{2\mu(1 + \lambda^2)}$$

$$\cos \alpha = \frac{3\lambda^2\mu^2 + \lambda^2 + \mu^2 - 1}{2\lambda\mu\sqrt{2(1 + \lambda^2)(1 + \mu^2)}},$$

$$\cos \beta = \frac{\lambda^2\mu^2 - \lambda^2 + 3\mu^2 + 1}{2\mu\sqrt{2(1 + \lambda^2)(1 + \mu^2)}}$$

Theorem

If the ratios $\frac{\|X_u\|}{\|X_v\|}$ and $\frac{\|Y_v\|}{\|Y_u\|}$ are constant, $X(u, v)$ is **isogonal (I-net)**, and $X(u, v)$ and $Y(u, v)$ form an **isogonal 4-web (I-web)**.

Discrete Isogonal Nets and Webs

Quad mesh $f: \mathbb{Z}^2 \rightarrow \mathbb{R}^3$:

vertices $v_i (1, \dots, 4)$ in each quad face

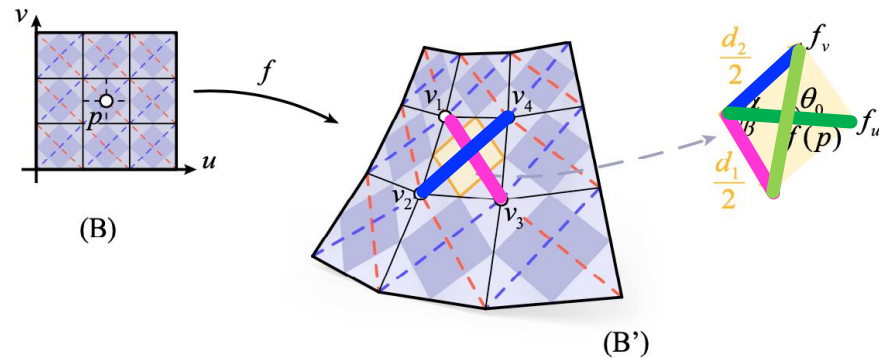
Discrete first-order smooth derivatives :

$$d_1 = v_3 - v_1, d_2 = v_4 - v_2$$

Discrete second-order smooth derivatives :

$$f_u = \frac{1}{2}(v_3 + v_4 - v_1 - v_2), f_v = \frac{1}{2}(v_1 + v_4 - v_2 - v_3)$$

$$\text{constants } \frac{\|d_2\|}{\|d_1\|} := \lambda, \frac{\|f_u\|}{\|f_v\|} := \mu \quad \Rightarrow \quad \theta(\lambda, \mu), \theta_0(\lambda, \mu), \alpha(\lambda, \mu), \beta(\lambda, \mu)$$



Theorem

If the diagonal ratio $\frac{\|d_2\|}{\|d_1\|}$ and medial-line ratio $\frac{\|f_u\|}{\|f_v\|}$ are constant, f is **isogonal (DI-net)**, and f and its diagonal quad mesh d form a discrete **isogonal 4-web (DI-web)**.

Special cases

- $\lambda = 1 \Leftrightarrow \theta_0 = \frac{\pi}{2} \Leftrightarrow X_u \perp X_v$

Definition. An I-net is an orthogonal net (**O-net**) if $\lambda=1$.

Definition. A DI-net is orthogonal (**DO-net**) if $\lambda=1$.

Corollary. A DO-net consists of similar **rhombuses** in the CBP.

- $\mu = 1 \Leftrightarrow \theta = \frac{\pi}{2} \Leftrightarrow Y_u \perp Y_v$

Corollary. A DI-net with $\mu=1$ consists of similar **rectangles** in the CBP.

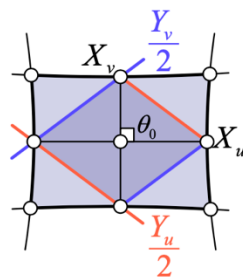
- $\lambda = \mu \Leftrightarrow \theta = \theta_0 \Leftrightarrow \frac{\|X_u\|}{\|X_u\|} = \frac{\|X_u+X_v\|}{\|X_u-X_v\|}$

Corollary. A DI-net with θ -Quads consists of similar **θ -Parallelograms** in the CBP.

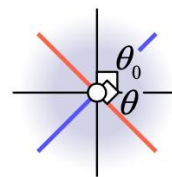
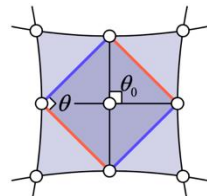
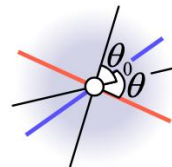
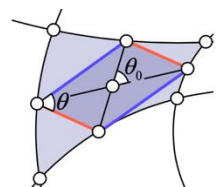
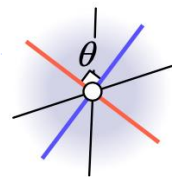
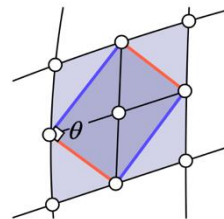
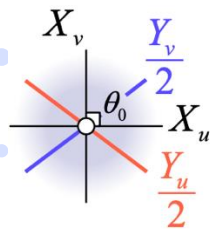
- $\lambda = \mu = 1 \Leftrightarrow \theta = \theta_0 = \frac{\pi}{2} \Leftrightarrow X_u \perp X_v, Y_u \perp Y_v$

Corollary. A DO-net with θ -Quads consists of **squares** in the CBP.

I-net



I-web



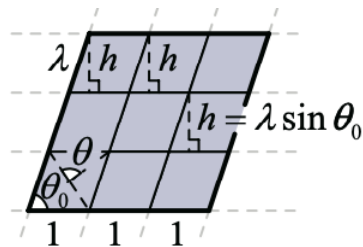
Construction of I-nets

- I-nets on planar sheared grid
- I-nets on developable surfaces
- I-nets from isothermal parametrizations
- I-nets by computational design

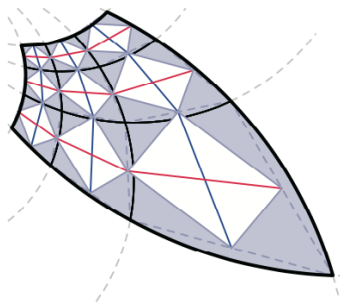


Construction of I-nets

- I-nets on planar sheared grid



$$\omega = \frac{1}{z}$$



$$C_1 = \{z = t + i\mathcal{A}, \frac{\mathcal{A}}{h} \subseteq \mathbb{Z}, 0 \notin \mathcal{A}, h \in \mathbb{R}\}$$

$$C_2 = \{z = t + \mathcal{B} + i(kt), k = \tan \theta \in \mathbb{R}, \mathcal{B} \subseteq \mathbb{Z}, 0 \notin \mathcal{B}\}$$

$$\omega = \frac{1}{z}$$

$$u + iv = \frac{t}{t^2 + \mathcal{A}^2} - i \frac{\mathcal{A}}{t^2 + \mathcal{A}^2}$$

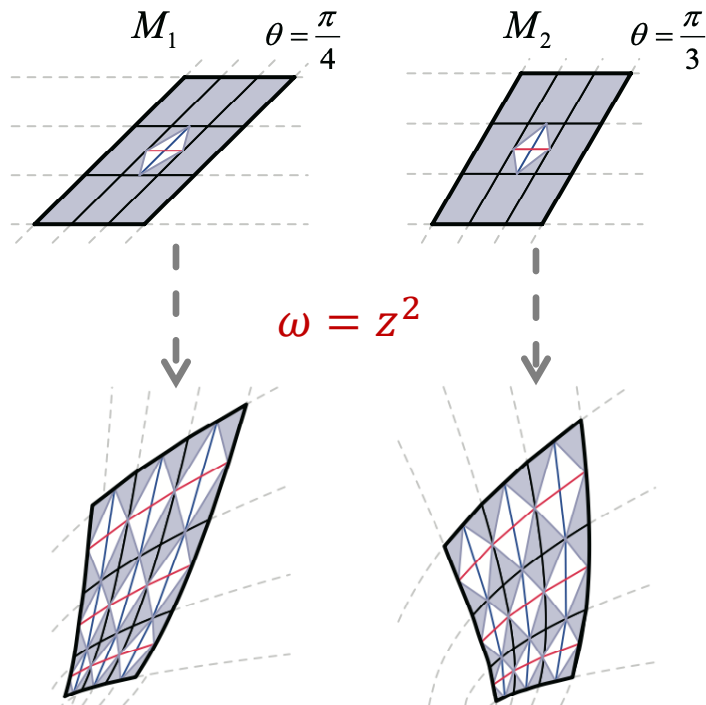
$$u + iv = \frac{t + \mathcal{B}}{(1 + k^2)t^2 + 2\mathcal{B}t + \mathcal{B}^2} - i \frac{kt}{(1 + k^2)t^2 + 2\mathcal{B}t + \mathcal{B}^2}$$



$$u^2 + (v + \frac{1}{2\mathcal{A}})^2 = \frac{1}{4\mathcal{A}^2}, (u - \frac{1}{2\mathcal{B}})^2 + (v - \frac{1}{2k\mathcal{B}})^2 = \frac{1 + k^2}{4k^2\mathcal{B}^2}$$

Construction of I-nets

- I-nets on planar sheared grid



$$C_1 = \{z = t + i\mathcal{A}, \frac{\mathcal{A}}{h} \subseteq \mathbb{Z}, 0 \notin \mathcal{A}, h \in \mathbb{R}\}$$

$$C_2 = \{z = t + \mathcal{B} + i(kt), k = \tan \theta \in \mathbb{R}, \mathcal{B} \subseteq \mathbb{Z}, 0 \notin \mathcal{B}\}$$

$$\omega = z^2$$

$$u + iv = t^2 - \mathcal{A}^2 + i(2\mathcal{A}t)$$

$$u + iv = (1 - k^2)t^2 + 2\mathcal{B}t + \mathcal{B}^2 + i(2kt^2 + 2k\mathcal{B}t)$$

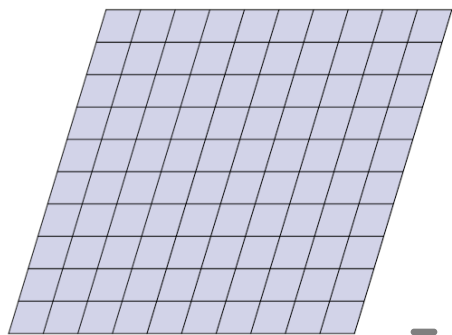
$$v^2 - 4\mathcal{A}^2u - 4\mathcal{A}^4 = 0$$

$$4k^2u^2 + (k^2 - 1)^2v^2 + 4k(k^2 - 1)uv +$$

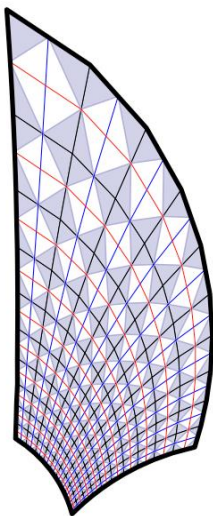
$$4k^2\mathcal{B}^2(k^2 - 1)u - 8k^3\mathcal{B}^2v - 4k^4\mathcal{B}^4 = 0.$$

Construction of I-nets

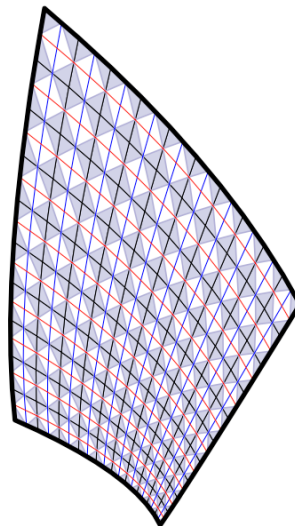
- I-nets on planar sheared grid



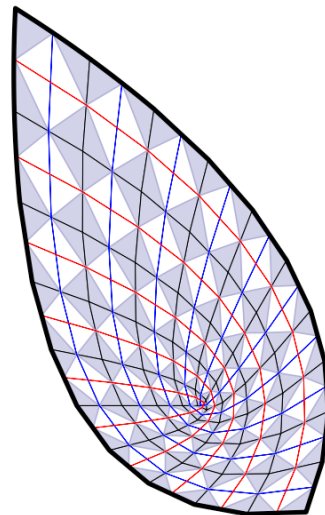
10 x 10 grid



$$\omega = \frac{1}{z}$$



$$\omega = z^2$$

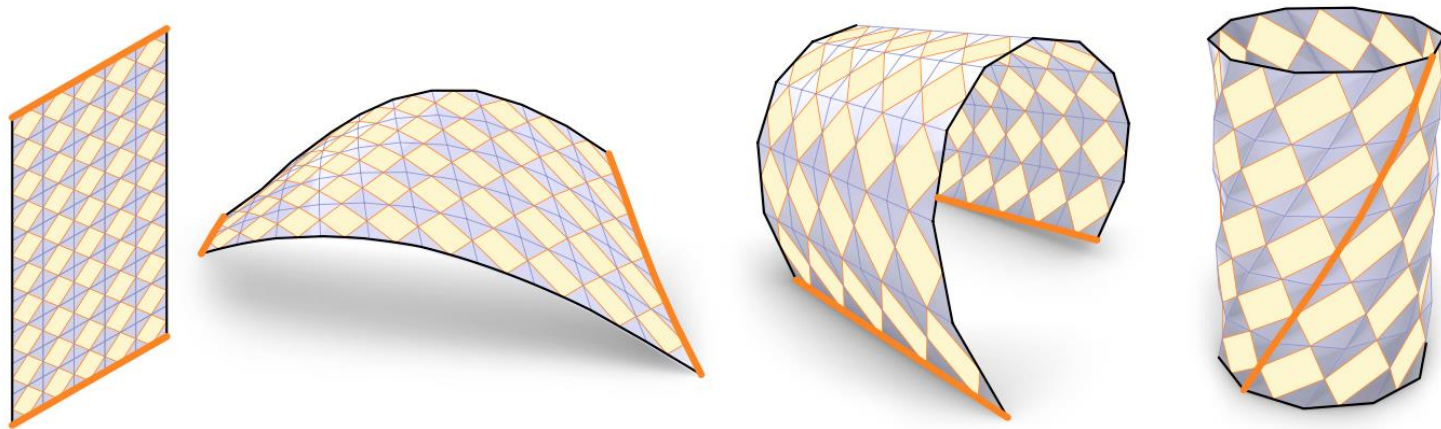


$$\omega = z^2$$

(Origin lies within the pre-image)

Construction of I-nets

- I-nets on developable surfaces



developable I-nets through isometric deformations

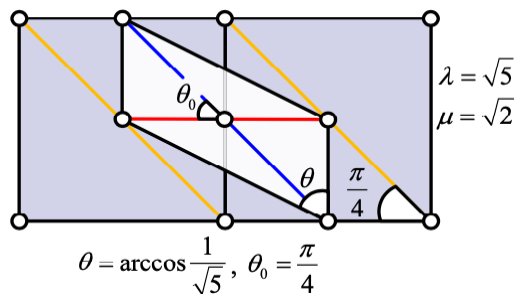


Construction of I-nets

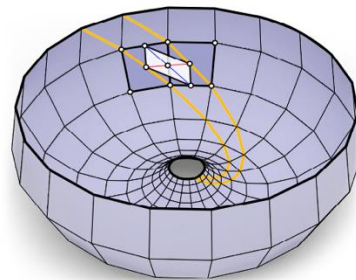
- I-nets from isothermal parametrizations

Isothermal parametrization $S(u, v)$:

- $F = S_u \cdot S_v = 0$, $E = S_u^2 = G = S_v^2$
- Conformal
- Maps **small squares** from domain to **small squares** on the surface



$$(\theta, \theta_0) = (\arccos \frac{1}{\sqrt{5}}, 45^\circ)$$



Construction of I-nets

- I-nets from isothermal parametrizations

Surface of revolution: $S(u, v) = (f(v)\cos u, f(v)\sin u, g(v)) \Rightarrow F = 0$

$$E = f^2(v) = f'^2(v) + g'^2(v) = G$$

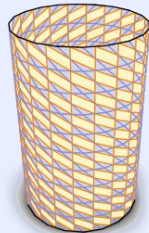
↓
Isothermal parametrizations

- Cylinder: $f(v) = r, g(v) = r v$
- Sphere: $f(v) = r \operatorname{sech} v, g(v) = r \tanh v$
- Catenoid (C): $f(v) = r \cosh v, g(v) = r v$
- Helicoid (H): $S(u, v) = (r \sinh v \cos u, r \sinh v \sin u, ru)$
- Conjugate minimal surface: $(\cos t) C + (\sin t) H, t \in \mathbb{R}$

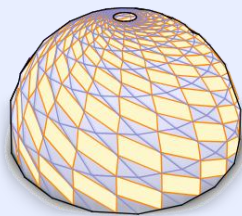


Isothermal net:

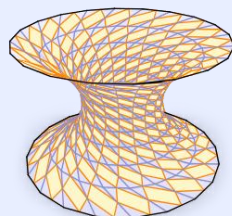
$$(\theta, \theta_0) = (\arccos \frac{1}{\sqrt{5}}, 45^\circ)$$



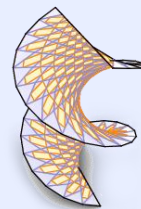
(A₁)



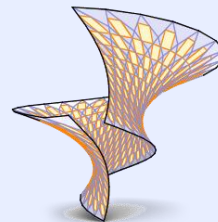
(B₁)



(C₁)

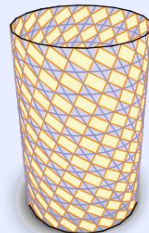


(D₁)

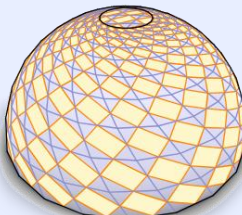


(E₁)

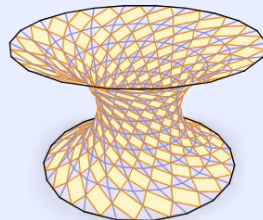
$$(\theta, \theta_0) = (80^\circ, 45^\circ) :$$



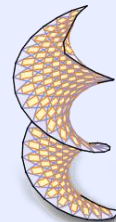
(A₂)



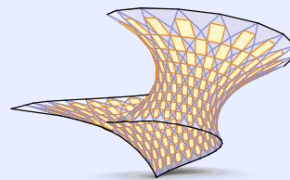
(B₂)



(C₂)



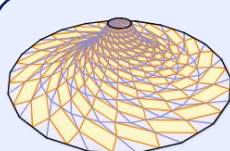
(D₂)



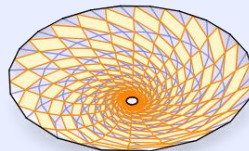
(E₂)

Möbius transf. :

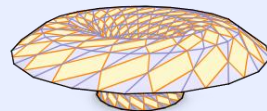
$$(\theta, \theta_0) = (\arccos \frac{1}{\sqrt{5}}, 45^\circ)$$



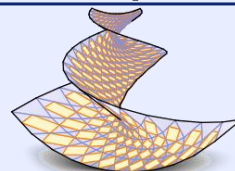
(A₃)



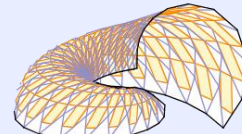
(B₃)



(C₃)

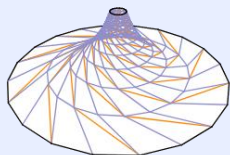


(D₃)

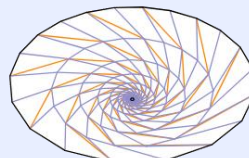


(E₃)

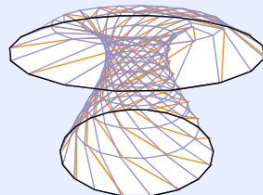
$$(\theta, \theta_0) = (50^\circ, 50^\circ) :$$



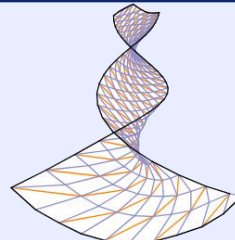
(A₄)



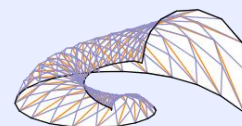
(B₄)



(C₄)



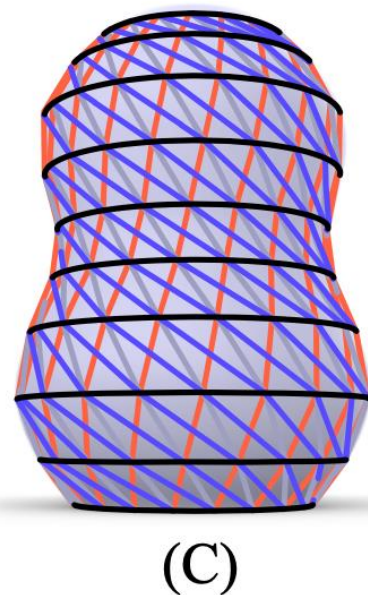
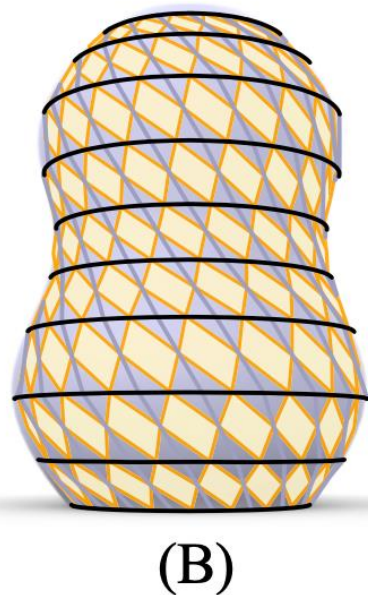
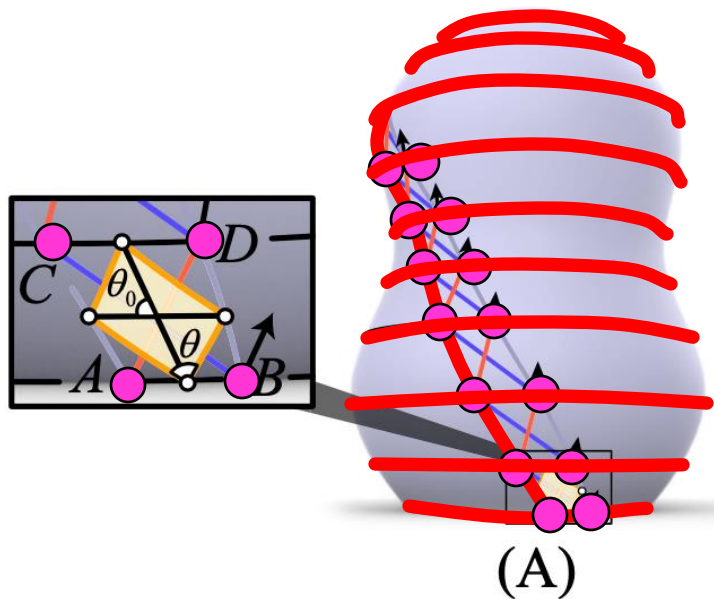
(D₄)



(E₄)

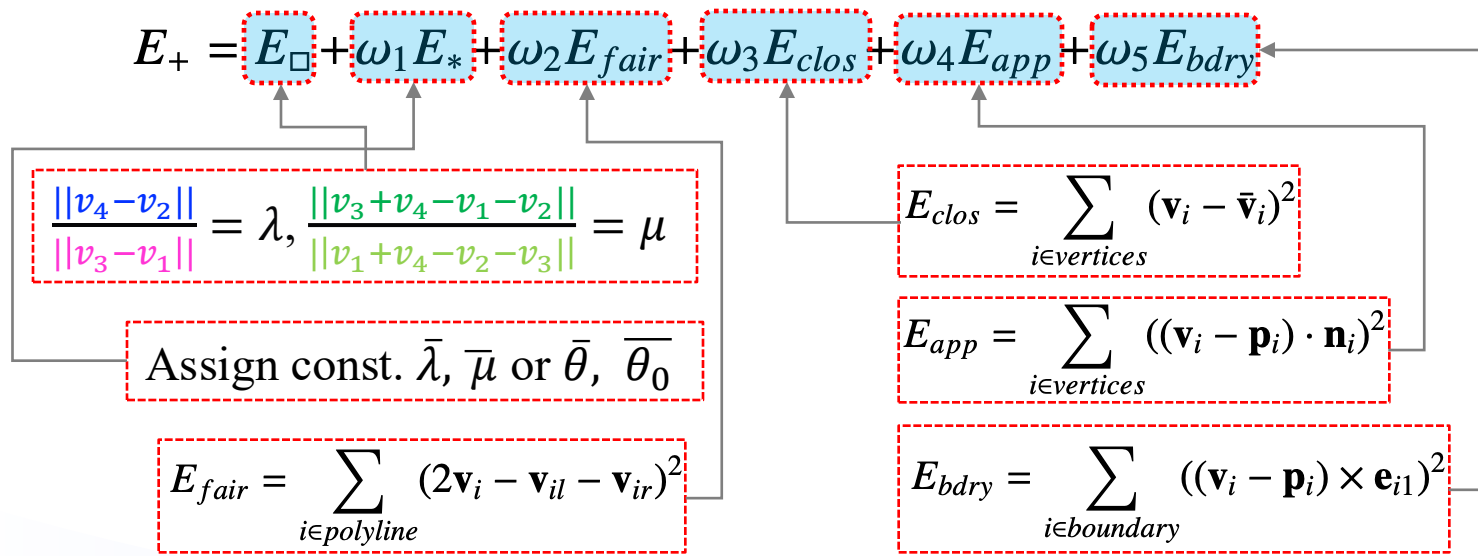
Construction of I-nets

- I-nets formed by loxodromes and parallels



Computational Design

- Initialization of I-nets
- Optimization



ArchGeo — Geometry Processing Library



Github Page

Geometric Visualization

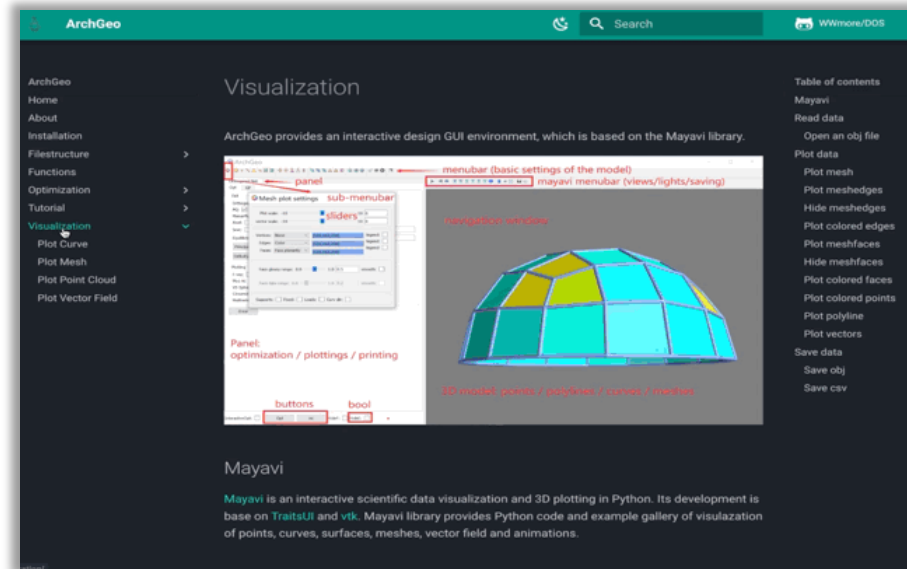
Point cloud, Polylines, Bezier / B-spline / NURBS curves / surfaces, Meshes, Curvatures, Vector fields...

Geometric Optimization

Curvature nets, Gridshell structures, Paneling, Parametrizations, Interactive editing...

```
1 # -*- coding: utf-8 -*-
2
3 Created on Sun Dec 18 22:16:21 2022
4
5 @author: WANGHUI
6
7 # All rights reserved.
8
9 # This software is free for non-commercial, research and evaluation use
10 # under the terms of the LICENSE.md file.
11
12 #
13 # For inquiries contact: hui.wang@xjtu.edu.cn
14
15 #
16 #
17
18 import os
19
20 os.environ['OMP_DUPLICATE_FIRST_ON'] = 'True'
21
22 import sys
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24 path = os.path.dirname(os.path.abspath(__file__))
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26 sys.path.append(path)
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```

Run readfile_orthonet.py



ArchGeo Window

<https://github.com/WWmore/DOS>

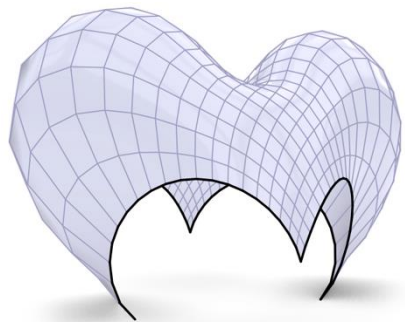
ArchGeo Documentation Page

<https://www.huiwang.me/mkdocs-archgeo/>

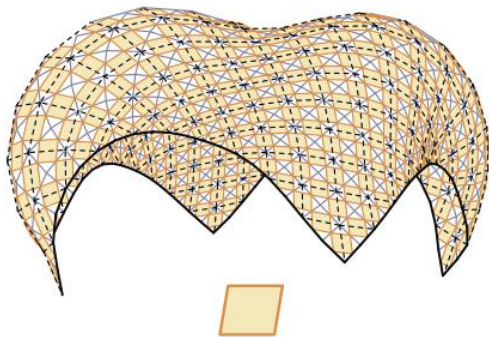


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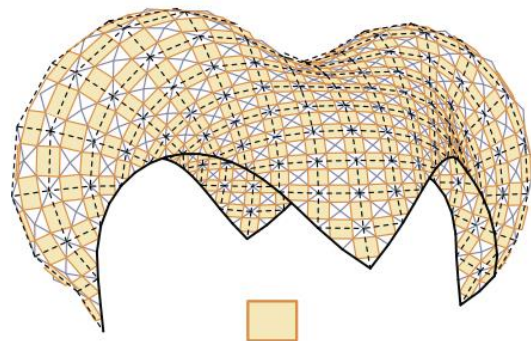
Results



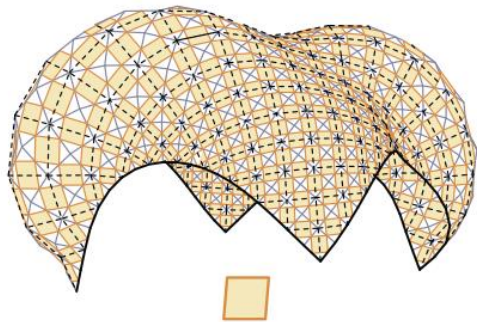
Initial mesh



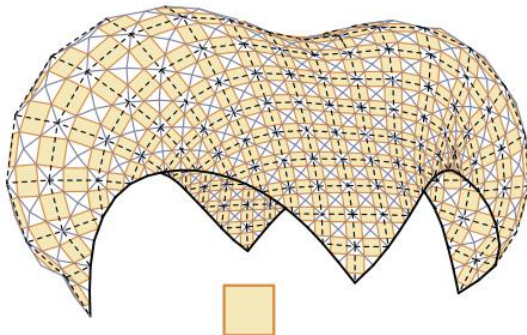
(A) $(\theta, \theta_0) = (80.42^\circ, 84.30^\circ)$



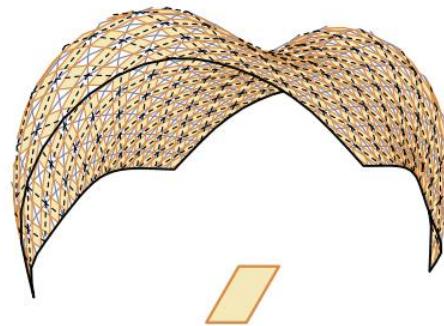
(B) $(\theta, \theta_0) = (90^\circ, 77.92^\circ)$



(C) $(\theta, \theta_0) = (85.27^\circ, 90^\circ)$

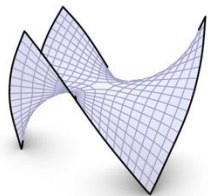


(D) $(\theta, \theta_0) = (90^\circ, 90^\circ)$

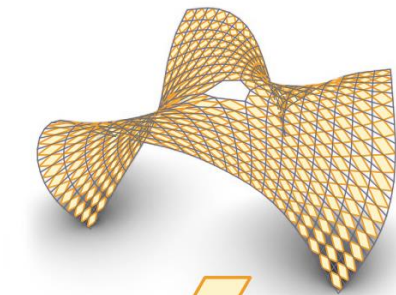
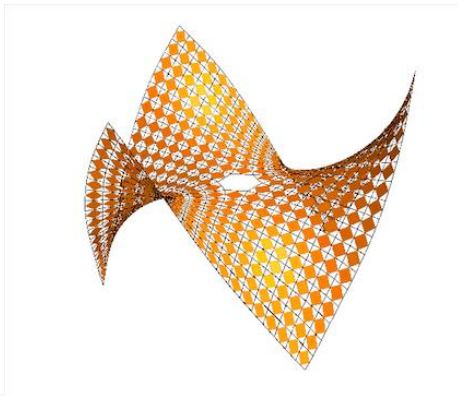


(E) $(\theta, \theta_0) = (60^\circ, 60^\circ)$

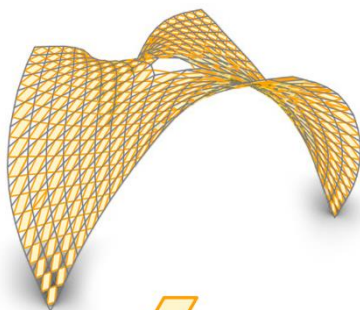
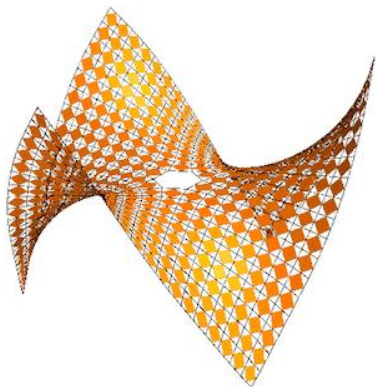
Results



Initial mesh

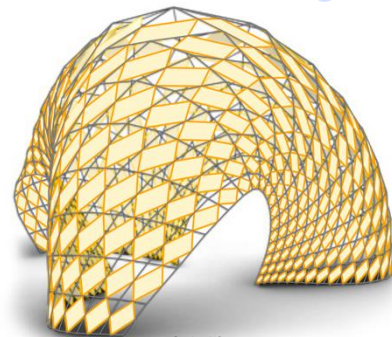


(A) $(\theta, \theta_0) = (60^\circ, 60^\circ)$



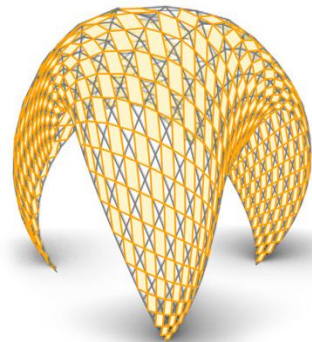
(B) $(\theta, \theta_0) = (60^\circ, 45^\circ)$

Mobius Transf.



(A')

Mobius Transf.



(B')

Conclusion

- A novel method for I-net and I-web using similar mid-edge subdivided parallelograms
- Expand the scope of geometric design possibilities and ensures compatibility with orthogonal nets
- Simplify the optimization process by constraining edge ratios instead of angles



Future work

- Integrate with other specialized nets, e.g. Weingarten surfaces
- Potential applications:
 - panelling design for freeform architectural surfaces
 - gridshell structures with uniform knots
 - repetitive pattern design
 - auxetic quads with similar elements
 - periodic structures in additive manufacturing
 - woven fabrics with regular patterns
 -



Thank you !

huiwang@xjtu.edu.cn



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同濟大學
TONGJI UNIVERSITY



RMIT
UNIVERSITY

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